

Convolution vs. Correlation

- Very similar methods
- Convolution is used for filtering image
- Correlation is used to detect patterns in an image
- They require a matrix called window or kernel, which defines the operation
- Convolution indexes filtering window in reverse, x-max is x-min, same for y axis.
- Most kernels are symmetric, thus have no difference between two methods
- The kernel is generally smaller than the image and has an odd size (e.g., 3x3)
- We will be working with convolution

Applying convolution

- In order to apply convolution, we need to visit every single pixel in the image
- Center the matrix on the pixel, multiply each value and sum the results. This is the new value for the pixel
- Do not replace the values, build a new image.
- If you are keen on maths, following is the formula for a kernel ω of size $n \times m$

$$O_{x,y} = \sum_{i=-\lfloor n/2 \rfloor}^{\lfloor n/2 \rfloor} \sum_{j=-\lfloor m/2 \rfloor}^{\lfloor m/2 \rfloor} \omega_{i+\lfloor n/2 \rfloor, j+\lfloor m/2 \rfloor} I_{x+i, y+j}$$

- The above formula will require values outside the image area. There are methods to handle them and we will discuss them later on. For now, ignore border pixels.
- Mirror the matrix so that first value becomes the last.
- Lets assume the following kernel is given

```
a  b  c
d  e  f
g  h  i
```

- By mirroring becomes

```
i  h  g
f  e  d
c  b  a
```

- Imagine we have an large image, but to demonstrate lets only consider the following segment by applying the kernel above to the pixel **q**

.	.	.	.	.
.	x	y	z	.
.	p	q	r	.
.	i	j	k	.
.	.	.	.	.

- Then our final value for the pixel can be calculated as follows:

$$ix + hy + zg + pf + rd + ic + jb + ka$$

- As you may have noticed, sum of the kernel values should be 1 in order to keep the brightness of the image fixed

Uses

- Convolution is used to apply various effects on the image
- These could be to reduce the effects of noise, enhance the quality of the image, or transform the image to be used in computer vision
- In addition to common uses above, convolution could be used to fake wind or motion effects

Dealing with edges

- Edges can cause invalid pixel locations to be required. We can deal with them in the following manner
- **Closest pixel value:** simply get the value of the pixel closest to the desired pixel. For instance, if you need the value for (-2, 1). Simply return the value for (0, 1). This method is also called clamping as it keeps the values in range. This is a good method for regular photos
- **Fixed color:** return a fixed color for the edges. This method is useful for computer generated images where there is a fixed canvas color (like white in this document). This will cause problems with photos
- **Cyclic:** act as if the image is tiled indefinitely. This is good for textures that tile well. If you need the value of (-2, 1) in an image of size w, return the value for (w-2, 1).
- **Mirror:** act as if image is tiled with mirroring. This is good for images that have continuous features, like the picture of a road. This is quite easy, take the absolute value of the pixel, (-2, 1) becomes (2, 1).

Types

- There are 3 main types of convolution filters
 - Low pass / Blur: softens image features, good to remove noise, makes images fuzzy. Sum of the kernel should be normalized to 1. Generally does not include negative values. If it does it will probably be in the center.
 - High pass / Sharpen: sharpens image features, sharper images generally looks more pleasing. Amplifies noise. Sum of the kernel should be normalized to 1. Generally contains many negative values and the center is a large positive value.
 - Band pass / Edge detection: generally used for computer vision to extract edges in the image. Sum of the kernel is 0 to remove DC component (average value of the image).

- Box filter: this is a filter where every value is equal. There might be exceptions to this rule to change filter type.
 - For low pass: use $\frac{1}{n \times m}$ for a kernel that is $n \times m$
 - For high pass: use -1 for every cell except for the center. The center should be $n \times m$
 - For edge detection: use -1 for every cell except for the center. The center should be $n \times m - 1$
 - Box filter is can cause artifacts in most cases and is not recommended. But due to its simplicity, it is used cases where quality is not important
- Gaussian filter
 - Uses weights from Gaussian distribution. Causes minimal artifacts.
 - Gaussian filter in convolution is Gaussian filter in frequency domain (we will talk about this later on).
 - Gaussian filter is used for blur. But same strategy that we used in box filter can be used here too (with different weights).
- Robert's cross: is a simple and effective edge detector. Do not center, use top left as reference point. It is fixed as follows:

$$\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}$$

- Sobel: is used to calculate image gradient, a very useful analysis tool. Should be calculated separately x and y directions:

$$G_x = \begin{array}{ccc} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{array} \quad G_y = \begin{array}{ccc} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{array}$$

Dealing with out of range values

- Sharpen filters can cause negative or out of range values
- Edge detection can cause negative pixel values
- There are multiple methods to handle these
 - Clamp the value: This is preferred method for sharpen. Negatives should become 0, out of range values should become highest possible value.
 - Contrast fitting: similar to contrast stretching, ensure all values fits in the range. This requires a temporary image that has more bit depth then the original. Generally used in edge detection (rarely in sharpen) and the stretch values are assumed to be $-(2^B - 1)$ and $(2^B - 1)$. If this can be assumed, temporary image might not be necessary
 - Mirror: Used in edge detection. Simply get the absolute value. Edges loses direction (in-out).